

Research paper

Awakening dormant nodes for robust long-term traffic forecasting

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ABSTRACT

A long-standing challenge in traffic forecasting is the presence of *dormant nodes*—sensors with persistently low or near-zero traffic. These nodes introduce low-quality, information-poor signals that can corrupt the learning process of deep learning models and degrade their ability to capture robust spatio-temporal patterns, particularly over long horizons. To address this, we propose a practical and effective feature engineering approach: augmenting the model input with a dedicated feature channel that encodes **prior knowledge about node dormancy**. This method requires no architectural changes and is readily pluggable into existing models. Building on this concept, we further develop the **Dormant Feature Embedding (DFE)**, a learnable module that transforms this static knowledge into an adaptive, task-specific representation. DFE enables the model to dynamically mitigate the negative influence of dormant nodes during training. Experiments demonstrate that our approach consistently improves both accuracy and efficiency. On high-dormancy datasets, DFE reduces the **Mean Absolute Percentage Error (MAPE)** by up to **17.3%**. Even on datasets with milder dormancy, it achieves up to an **11.15%** improvement in long-term forecasting scenarios, while reducing training time to approximately **75%** of state-of-the-art baselines.

1. Introduction

1.1. Research background

Accurate traffic flow forecasting is essential for smart-city development, supporting better mobility, congestion control, and urban planning. With widespread sensor deployment, urban road networks now generate abundant spatiotemporal data, enabling the discovery of latent patterns and improved predictive accuracy (Zhang et al., 2017a; Cirstea et al., 2021).

In recent years, deep-learning methods have progressively improved traffic forecasting accuracy and extended prediction horizons. These models typically combine recurrent temporal modeling with graph- or attention-based spatial mechanisms to capture complex dependencies in the data (Zhao et al., 2020; Yu et al., 2018; Zhang et al., 2017b; Greff et al., 2017; Duan et al., 2016; Park et al., 2020; Liu et al., 2017; Han et al., 2019a; Xue et al., 2023; Zhang et al., 2023). Significant efforts have also been made to extend long-range forecasting capabilities (Zhang et al., 2024; Guo et al., 2023).

1.2. Problem statement

A persistent challenge in traffic forecasting is the data imbalance caused by the network structure itself. While some nodes (sensors) on

major thoroughfares are consistently active, a significant number of nodes remain persistently inactive or exhibit near-zero traffic flow. We refer to these as *dormant nodes*.

While traffic networks are widely known to be sparse, we observe a subset of nodes that reside in an extreme state of inactivity—remaining at near-zero flow for days. Unlike general low-traffic nodes that may still carry meaningful temporal variations, these “dormant nodes” provide near-constant, information-poor signals that hinder the capture of robust spatio-temporal patterns.

As illustrated in Fig. 1, these nodes produce nearly flat time series, especially after normalization, which results in almost constant input values for the model.

This data characteristic becomes particularly detrimental in long-term forecasting tasks. The near-constant inputs from dormant nodes are information-poor, providing an unreliable and noisy basis for learning. This corrupts the feature aggregation process in spatio-temporal models, as active nodes are forced to incorporate these noisy signals. Ultimately, this degrades the model’s ability to learn robust and coherent global dynamics across the entire network.

1.3. Proposed solution overview

To mitigate the adverse impact of dormant nodes, we introduce an intuitive yet effective strategy tailored for spatially coupled networks:

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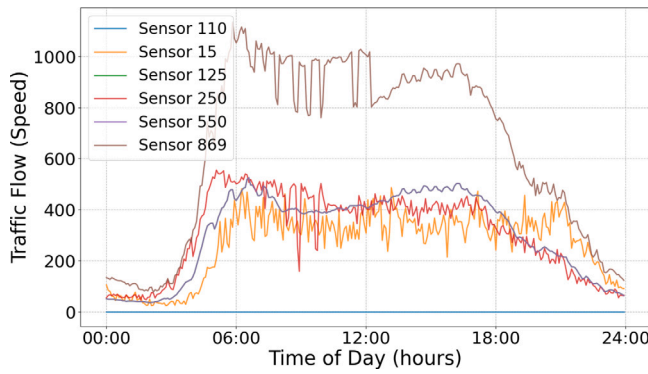


Fig. 1. PEMS07 traffic flow time series. Sensors 869 exhibit high traffic activity and pronounced temporal variations throughout the day, while sensors 110 maintain consistently low or near-zero traffic flow, qualifying them as dormant nodes.

explicitly augmenting the model's input with a feature that highlights these structurally inactive nodes. We first propose a simple realization of this idea, the Dormant Feature Indicator (DFI)—a binary, non-learnable mask derived from historical traffic statistics. DFI serves as a sanity-check baseline and a reference for ablation studies.

Building upon this concept, we develop the Dormant Feature Embedding (DFE)—a lightweight, input-level enhancement. DFE adds a learnable feature channel, initialized from historical traffic patterns, to the input data. This embedding guides the model to better utilize information from persistently low-activity nodes and improves model convergence, especially in long-term forecasting scenarios. An overview of the DFE framework is presented in Fig. 2.

This additional feature serves two primary purposes. First, it acts as a stable, informative guiding signal, allowing the model to recognize the unreliable nature of the primary traffic data from these nodes and adapt its learning strategy accordingly. Second, it injects valuable spatial priors regarding node importance and activity patterns into the model. These enhancements help the model better capture global spatio-temporal dependencies, significantly boosting long- and ultra-long-term forecasting accuracy.

1.4. Contributions

Our main contributions are fourfold:

- We identify and analyze the problem of dormant nodes — sensors with persistently low or near-zero traffic — as a key and previously overlooked factor that hinders spatio-temporal learning, especially in long-horizon forecasting tasks.
- We propose a novel feature augmentation strategy to mitigate this issue, introducing a simple Dormant Feature Indicator (DFI) and its more advanced, learnable variant, the Dormant Feature Embedding (DFE).
- We provide a theoretical analysis, grounded in information theory, that explains how dormant nodes degrade signal quality and corrupt the learning process, how our feature augmentation provides a guiding signal to mitigate this issue, and why the DFE design is well-justified.
- Through extensive experiments, we demonstrate that introducing dormant features significantly improves forecasting accuracy and training efficiency across multiple models and datasets, particularly under long- and ultra-long-term settings and in scenarios with a severe presence of dormant nodes.

2. Related work

In this section, we briefly review early approaches to traffic forecasting, followed by short-term prediction methods based on spatial-temporal deep neural networks, and conclude with an overview of the latest neural architectures designed for long-term and ultra-long-term traffic flow prediction.

2.1. Early studies for traffic forecasting

Early traffic forecasting predominantly relied on conventional statistical techniques, including Historical Average (HA) (Box et al., 2015), Kalman Filters (KF), and various ARIMA models (Lee and Fambro, 1999; Williams and Hoel, 2003). While straightforward, these methods struggle to capture the complex non-linear characteristics inherent in traffic data. Concurrently, traditional machine learning approaches were explored, such as time-aware nearest neighbors (Dell'Acqua et al., 2015) and Fourier-SVR hybrids (Luo et al., 2019). However, their heavy reliance on extensive manual feature engineering, limited parameter capacities, and suboptimal computational efficiency ultimately restrict their practical applications.

2.2. Spatial-temporal neural networks for traffic forecasting

In recent years, deep learning approaches have gained substantial adoption in traffic forecasting. Early sequence-to-sequence models primarily focused on extracting temporal features, with works like Duan et al. applying LSTM (Duan et al., 2016), and Fu et al. leveraging both LSTM and GRU for traffic time series (Fu et al., 2016).

To incorporate the spatial attributes of road networks, convolutional neural networks (CNNs) were introduced. ST-ResNet extracts geometric features from grid-segmented networks (Zhang et al., 2017b), while related frameworks integrate CNN-based deep-clustering (Han et al., 2019b) or combine CNNs with LSTMs to capture multi-scale dependencies (Wu and Tan, 2016). However, the non-Euclidean nature of real-world road networks poses significant challenges for standard CNNs.

Consequently, graph neural networks (GNNs) — widely successful in diverse domains (Chen et al., 2024; Qin et al., 2024; Xi et al., 2024) — have become the standard for traffic modeling. T-GCN (Zhao et al., 2020) and STGCN (Yu et al., 2018) effectively combine static graph convolutions with temporal modules to capture spatio-temporal dependencies. STMGCN further extends this by capturing inter-regional correlations from multi-graph perspectives (Geng et al., 2019). To overcome the limitations of manually predefined graphs, Graph WaveNet (GWN) introduced adaptive spatial correlations via learnable embedding vectors (Wu et al., 2019).

Addressing time-varying spatial correlations, models like DMSTGCN (Han et al., 2021) and DSTAGNN (Lan et al., 2022) generate data-driven, dynamic spatio-temporal graphs. Similarly, Guo et al. utilized self-attention to dynamically adjust static weight matrices (Guo et al., 2022). Parallely, attention mechanisms (T. Zhang et al., 2024) have been adapted to dynamically focus on salient spatial-temporal features. ASTGCN integrates attention with graph convolutions (Guo et al., 2019), STRGAT utilizes residual graph-attention blocks (Zhang et al., 2023), and MISTAGCN fuses multi-input attention with standard convolutions (Tao et al., 2023). Additionally, Fourier-series embeddings have been combined with attention and multi-hop diffusion to address large-scale structural dependencies (Xue et al., 2023).

While these advanced methodologies improve short-term prediction, their complex architectures and heightened computational resource requirements severely constrain their applicability in long-term forecasting scenarios.

Consequently, the evolution of spatial-temporal networks for long-term forecasting has focused on efficiently capturing extended dependencies. Early works like AutoST (Li et al., 2022) automated spatio-temporal feature extraction. SSTBAN (Guo et al., 2023) pioneered

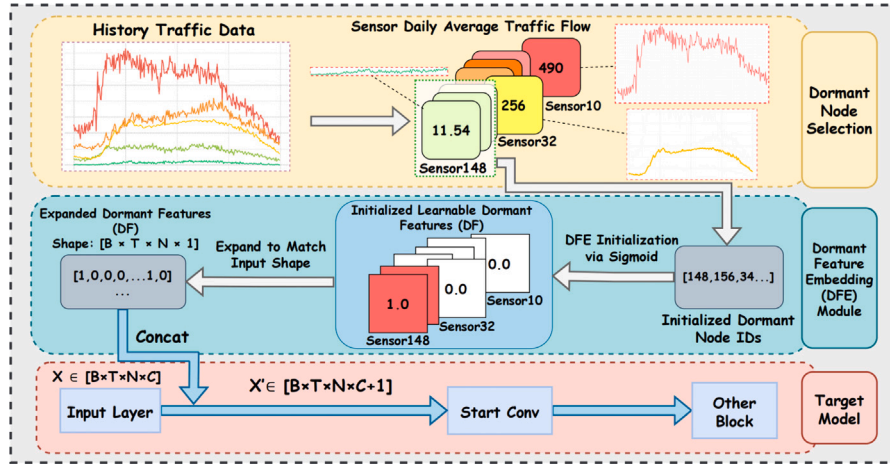


Fig. 2. Overview of the DFE framework. The color coding across the signal curves and sensor blocks visually represents traffic activity levels, ranging from high flow (red) to extreme dormancy (green, e.g., Sensor 148). The proposed Dormant Feature Embedding (DFE) is injected as an additional feature channel at the input layer, allowing any downstream spatio-temporal model to leverage information about structurally inactive nodes to enhance long-term forecasting performance.

deep-learning-based long-term traffic forecasting using a bottleneck-attention scheme to reduce computational cost to $\mathcal{O}(TN)$. Most recently, Extralonger (Z. Zhang et al., 2024) has pushed the envelope further by introducing efficient long-dependency modeling strategies that extend the prediction horizon to over one week.

However, it is noteworthy that most previous work has largely overlooked the significant impact that dormant nodes have on traffic flow forecasting, particularly for long- and ultra-long-term predictions. In contrast to earlier studies, our work is one of the first to explicitly identify and substantiate that dormant nodes represent a critical issue that can severely degrade prediction performance over extended horizons. To mitigate this challenge, we introduce a plug-and-play feature augmentation strategy, realized through our proposed Dormant Feature Indicator (DFI) and its learnable variant, the Dormant Feature Embedding (DFE). This approach effectively compensates for the lack of informative gradient signals from low-activity nodes. Our theoretical analysis and extensive experimental evaluations demonstrate that this strategy significantly alleviates the adverse effects of dormant nodes, thereby advancing the state-of-the-art in long-term traffic forecasting.

3. Problem formulation

In the context of traffic forecasting, we model the urban road network as a directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{A})$. Here, $\mathcal{V}$ represents the set of $N = |\mathcal{V}|$ nodes, which correspond to sensor locations deployed on the road network. $\mathcal{E}$ is the set of edges representing the road segments that connect these locations. The graph's topology and connectivity are captured by a weighted adjacency matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$, where each entry A_{ij} can represent spatial proximity, road distance, or other correlations between node i and node j .

The historical traffic data collected from the sensor network over a specific period is represented by a multi-dimensional tensor $\mathbf{X} = (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_{T_h}) \in \mathbb{R}^{T_h \times N \times C}$. In this tensor, T_h is the length of the historical time window, N is the number of nodes, and C is the number of features recorded at each node. These features typically include traffic speed and volume, and can be augmented with temporal metadata such as time-of-day or day-of-week indicators.

The task of spatio-temporal traffic forecasting can be formulated as learning a mapping function $\mathcal{F}(\cdot)$. This function takes the historical traffic data $\mathbf{X}$ and the graph structure $\mathcal{G}$ as input to predict the traffic conditions for the next T_p future time steps. The prediction is denoted by $\hat{\mathbf{Y}} = (\hat{\mathbf{x}}_{T_h+1}, \dots, \hat{\mathbf{x}}_{T_h+T_p}) \in \mathbb{R}^{T_p \times N \times C}$. The objective is formally expressed as:

$$[\mathbf{X}, \mathcal{G}] \xrightarrow{\mathcal{F}(\cdot|\Theta)} \hat{\mathbf{Y}}, \quad (1)$$

where Θ represents all the trainable parameters of the model $\mathcal{F}$. These parameters are optimized by minimizing a loss function $\mathcal{L}$ that measures the discrepancy between the prediction $\hat{\mathbf{Y}}$ and the ground truth observations $\mathbf{Y}$.

4. Methodology

As identified in the introduction, the presence of dormant nodes poses a significant challenge to learning effective spatio-temporal representations. The core of this problem is the degradation of signal quality from these nodes, which provides an unreliable, information-poor basis for learning. To address this, we propose a feature augmentation strategy that provides the model with explicit meta-information about node dormancy at the input level. We introduce two distinct implementations of this strategy: a simple, non-learnable indicator and a more flexible, learnable embedding.

4.1. Dormant Feature Indicator (DFI)

As a straightforward baseline to validate our hypothesis that explicitly signaling dormancy is beneficial, we first propose the Dormant Feature Indicator (DFI). DFI is a static, binary mask that identifies nodes with historically low traffic activity.

The construction of DFI involves a simple statistical analysis. First, we compute a dormancy score s_i for each node $i \in \mathcal{V}$, defined as its mean traffic value over a long-term historical period. Then, we identify the set of dormant nodes, $\mathcal{V}_d \subset \mathcal{V}$, by selecting a bottom percentile (e.g., 5%) of nodes with the lowest scores. The DFI is then formulated as a binary vector $\mathbf{m} \in \{0, 1\}^N$, where its i th element m_i is defined as:

$$m_i = \begin{cases} 1 & \text{if node } i \in \mathcal{V}_d \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

This indicator vector is broadcasted to match the input tensor's dimensions and concatenated along the feature channel, creating an augmented input $\mathbf{X}_{\text{DFI}} \in \mathbb{R}^{T_h \times N \times (C+1)}$. This process flags dormant nodes to the model without introducing any trainable parameters.

4.2. Dormant Feature Embedding (DFE)

While DFI provides a rigid, binary distinction, the boundary between dormant and active nodes can be ambiguous, and the optimal representation of dormancy might be task- or model-dependent. To

introduce greater flexibility and adaptivity, we propose the Dormant Feature Embedding (DFE), a learnable variant of our core idea.

DFE replaces the static binary indicator with a learnable embedding vector $\theta \in \mathbb{R}^N$. To maintain a connection to the statistical properties of the data, we initialize θ such that the initial embedding values for statistically identified dormant nodes are high (resulting in a value near 1 after activation), while those for active nodes are low.

To ensure the embedding remains near-binary while allowing for gradient-based optimization, we employ the Gumbel-Sigmoid reparameterization trick for sampling. This technique allows for differentiable sampling from a discrete distribution. The final dormant feature mask $\mathbf{m} \in [0, 1]^N$ is generated as follows:

$$\mathbf{m} = \sigma \left(\frac{\theta + \mathbf{g}}{\tau} \right), \quad (3)$$

where $\mathbf{g} \in \mathbb{R}^N$ is a vector of Gumbel noise samples, τ is a temperature parameter that controls the smoothness of the sampling, and $\sigma(\cdot)$ is the sigmoid function. The resulting soft mask $\mathbf{m}$ is then concatenated to the input features, yielding $\mathbf{X}_{\text{DFE}} \in \mathbb{R}^{T_h \times N \times (C+1)}$. This allows the model to fine-tune the importance of the dormancy signal for each node during end-to-end training.

4.3. Plug-and-play integration

A key advantage of both DFI and DFE is their design as lightweight, model-agnostic modules. They are applied only at the input layer, augmenting the feature representation without requiring any modifications to the architecture of the downstream spatio-temporal forecasting model. This plug-and-play characteristic ensures seamless integration into a wide variety of existing frameworks with negligible computational overhead, making our proposed methods a practical and versatile solution.

5. Theoretical motivations and analysis

In this section, we provide a principled analysis to conceptually ground the design and effectiveness of our proposed Dormant Feature Embedding (DFE). While the empirical results in Section 6 demonstrate its practical benefits, a deeper theoretical intuition is crucial for explaining *why* it works. To systematically elaborate on the motivations and mechanisms behind our approach, we structure this analysis around three critical questions:

- **(Q1) The Impact of Dormant Nodes:** Why and how do dormant nodes, which are prevalent in traffic datasets, fundamentally impair the learning process of spatio-temporal models?
- **(Q2) The Mechanism of Feature Augmentation:** How does our proposed input-level feature augmentation strategy effectively counteract the issues caused by dormant nodes.
- **(Q3) The Rationale for a Learnable Design:** What are the theoretical advantages of designing DFE as an adaptive, learnable embedding rather than a static, predefined indicator like DFI?

By addressing these questions, we aim to offer clear theoretical intuitions that complement our experimental findings and validate the architectural choices of the DFE module.

5.1. Q1: An information-theoretic view on the impairment mechanism

The detrimental impact of dormant nodes is best understood from an information-theoretic perspective, where their primary failure is a severe degradation of signal quality. This degradation not only makes the nodes themselves difficult to model but also pollutes the learning process across the entire network. This mechanism can be deconstructed into two key stages.

Stage 1: Input signal degradation via normalization. The process begins with the raw signal from a dormant node, which inherently possesses a low Signal-to-Noise Ratio (SNR). The “signal” is the meaningful, albeit subtle, variation in its traffic, while the “noise” includes sensor inaccuracies and random fluctuations. The common practice of global min–max normalization exacerbates this issue by squashing the node’s entire time series into a near-constant, near-zero sequence.

This compression can be formally characterized by Information Entropy. For a discrete random variable X with probability mass function $p(x)$, the entropy is defined as:

$$H(X) = - \sum_{x \in \mathcal{X}} p(x) \log_2 p(x) \quad (4)$$

A near-constant signal is highly predictable, meaning its probability distribution is concentrated on a single value. This results in extremely low entropy ($H(X) \rightarrow 0$), signifying a lack of information content. The normalization process, therefore, transforms an already low-SNR signal into an information-poor input for the model.

Stage 2: Corruption of spatio-temporal learning. This lack of information has profound consequences for learning network-wide dependencies, a process that relies on capturing the Mutual Information between nodes. The mutual information $I(X; Y)$ between two nodes’ signals, X and Y , quantifies their statistical dependency:

$$I(X; Y) = \sum_{y \in \mathcal{Y}} \sum_{x \in \mathcal{X}} p(x, y) \log_2 \left(\frac{p(x, y)}{p(x)p(y)} \right) \quad (5)$$

A fundamental property of mutual information is that it is bounded by the entropy of the individual variables, i.e., $I(X; Y) \leq \min(H(X), H(Y))$. Since a dormant node’s signal X_d has near-zero entropy after normalization ($H(X_d) \rightarrow 0$), the mutual information it can share with any neighboring node X_a is also forced towards zero ($I(X_d; X_a) \rightarrow 0$). Consequently, the model is theoretically incapable of learning meaningful relationships involving these dormant nodes.

This leads to two global learning impairments. First, during feature aggregation (e.g., message passing in GNNs), dormant nodes contribute what is essentially noise, corrupting the representations of their active neighbors. Second, it affects the optimization process itself. The back-propagated gradients originating from these information-poor signals are inherently weak. In the stochastic environment of model training, these weak “signal gradients” are easily overwhelmed by “gradient noise”—random fluctuations arising from mini-batch sampling and other sources of randomness. The resulting parameter updates become erratic and non-convergent, not because the gradient vanishes in the classical sense, but because the update signal is lost in the noise. This prevents the model from forming a stable, coherent understanding of the overall traffic network.

5.2. Q2: DFE as a guiding signal for robust learning

Our proposed feature augmentation strategy, DFE, is not designed to directly “fix” or “amplify” weak gradients. Instead, it alleviates the issues described in Q1 by providing the model with explicit, semantic meta-information. It fundamentally reframes the learning task: rather than forcing the model to interpret a corrupted, low-information signal, we provide it with a “warning label” that helps it recognize and adapt to the signal’s inherent unreliability. This mechanism can be understood through the lens of uncertainty reduction.

Reducing uncertainty with conditional information. As established in Q1, the normalized signal from a dormant node, let us call it X_{traffic} , is information-poor and noisy. For the model, predicting the true underlying traffic pattern, Y , based on this corrupted signal is a task fraught with high uncertainty. In information theory, this uncertainty can be quantified by Conditional Entropy, $H(Y|X)$, which measures the remaining uncertainty about Y when X is known:

$$H(Y|X) = - \sum_{x \in \mathcal{X}} p(x) \sum_{y \in \mathcal{Y}} p(y|x) \log_2 p(y|x) \quad (6)$$

For a dormant node, the conditional entropy $H(Y|X_{traffic})$ is high, because a single noisy, near-zero signal $X_{traffic}$ could correspond to many possible true underlying patterns Y . The model is essentially “confused” by the input.

Our DFE module introduces a new, clean feature channel, X_{DFE} , which explicitly indicates the dormancy status of the node. The model’s task is now to predict Y given both the traffic signal and the dormancy flag. The new uncertainty is $H(Y|X_{traffic}, X_{DFE})$. By providing this crucial context, we enable the model to disambiguate the signal. For instance, it can learn that when X_{DFE} is high, $X_{traffic}$ is unreliable. This additional information reduces the model’s confusion, leading to a fundamental inequality:

$$H(Y|X_{traffic}, X_{DFE}) < H(Y|X_{traffic}) \quad (7)$$

This demonstrates that DFE theoretically reduces the problem’s inherent uncertainty, making the learning task easier and more stable.

Enabling conditional processing and preserving structural roles. By reducing uncertainty, DFE empowers the model to learn a *conditional processing strategy*. A sufficiently complex neural network can learn to dynamically adjust its behavior based on the DFE signal. For example, it might learn to down-weight the influence of the ‘traffic’ feature from dormant nodes during spatial feature aggregation, thereby preventing noise from propagating through the network.

Furthermore, even if a node’s traffic data is uninformative, its existence and location within the graph — its structural role — remain critical for understanding global traffic dynamics. DFE provides a stable, persistent feature that clearly identifies these nodes. This ensures that dormant nodes are not simply ignored or treated as zero-vectors, but are recognized for their topological importance, allowing the model to preserve and utilize the complete graph structure for a more holistic understanding of the traffic network. In essence, DFE guides the model to distinguish between a node’s (unreliable) dynamic state and its (important) static role.

5.3. Q3: The rationale for a learnable and adaptive design

Having established that providing dormancy information is beneficial, we now justify the design of DFE as a learnable embedding rather than a static, heuristic-based indicator like DFI. The superiority of a learnable design is rooted in its enhanced expressiveness and its ability to adapt to the specific learning task through end-to-end optimization.

Beyond static heuristics: Expressiveness and nuance. A static indicator like DFI relies on a predefined, hard threshold (e.g., nodes with average traffic in the bottom 5%). This approach has two fundamental limitations. First, the threshold is arbitrary and likely suboptimal; the ideal definition of “dormancy” may vary across different datasets and even different parts of a single network. Second, it treats dormancy as a binary state (a node is either dormant or not). In reality, dormancy is a continuous spectrum: some nodes are perpetually inactive, while others are merely less active.

A learnable DFE, which outputs a continuous value for each node, possesses far greater expressiveness. It can capture the nuanced *degree* of dormancy, assigning different values to nodes with different activity levels. This allows the model to learn a more complex, fine-grained relationship between a node’s typical activity level and its reliability as an information source, moving beyond a coarse, one-size-fits-all heuristic.

An optimization perspective: Task-specific adaptation. The most significant advantage of DFE lies in its integration into the model’s optimization process. Let Θ represent the parameters of the main spatio-temporal model. The overall learning objective is to minimize a loss function L .

For a model using a static indicator M_{DFI} , the feature is a fixed constant. The optimization problem is solely over the model’s parameters:

$$\min_{\Theta} L(\text{Input} = X \oplus M_{DFI}; \Theta) \quad (8)$$

Here, $\oplus$ denotes feature concatenation. The model must learn to interpret a fixed, predefined signal.

In contrast, the DFE module introduces its own set of learnable parameters, ϕ , which generate the dormancy embedding $M_{DFE}(\phi)$. The optimization becomes a joint problem over both the model and the DFE parameters:

$$\min_{\Theta, \phi} L(\text{Input} = X \oplus M_{DFE}(\phi); \Theta) \quad (9)$$

This formulation allows the dormancy signal itself to be fine-tuned during training. Through backpropagation, the model can adjust the DFE embedding (ϕ) to represent dormancy in a way that is most beneficial for minimizing the specific loss function of the prediction task. This means the features and the model can co-adapt. For instance, for an ultra-long-term forecasting task, the model might learn that even moderately active nodes exhibit unreliable patterns and thus increase their corresponding DFE values.

In essence, DFE transforms the manual, heuristic-based process of defining dormancy into a principled, data-driven optimization problem, allowing the model to learn the optimal “guiding signal” for the task at hand. This adaptability makes DFE a more robust and powerful solution.

Comparison with structural and attention-level strategies. We also acknowledge that node reliability can be explicitly modeled at the structural level (e.g., by modifying the adjacency matrix) or through internal attention mechanisms. While such “intrusive” strategies might offer direct control for specific backbones, DFE prioritizes model-agnosticism and computational efficiency. By operating at the input level, DFE avoids the need for complex, model-specific architectural modifications, allowing it to serve as a universal “plug-and-play” enhancement for any spatio-temporal framework. Although this design relies on the downstream model’s capacity to interpret the augmented features, our experimental results across diverse architectures demonstrate that this lightweight guiding signal is sufficient to yield substantial performance gains without the overhead of structural interventions.

6. Experiments

In this section, we present a comprehensive empirical study to validate the theoretical underpinnings and practical utility of our proposed methods, DFI and DFE. Our experiments are meticulously designed to move beyond simple performance metrics and to provide a holistic understanding of the dormant node problem and our proposed solution. To this end, we structure our investigation around a series of key research questions (RQs), which are organized into four distinct categories:

- I. Core Performance Evaluation:** We begin by assessing the direct impact of our methods on forecasting accuracy across various prediction horizons.
 - (Q1) What is their performance impact on **long-horizon** prediction, where the dormant node problem is more pronounced?
 - (Q2) Can the improvements be sustained in challenging **extra-long-term** forecasting scenarios?
 - (Q3) How effective are DFI and DFE in standard **short-term** forecasting tasks?
- II. Practicality and Robustness:** We then evaluate the real-world viability and reliability of DFE under various conditions.
 - (Q4) What is the impact of DFE on computational resources, including memory usage, training time, and inference speed?

Table 1
Summary of datasets used in experiments.

Dataset	# Time steps	# Nodes	Time range
PEMS-03	26,208	358	2018/09–2018/11
PEMS-04	16,992	307	2018/01–2018/02
PEMS-07	28,224	883	2017/05–2017/08
PEMS-08	17,856	170	2016/07–2016/08
METR-LA	34,272	207	2012/03–2012/06
PEMS-BAY	52,116	325	2017/01–2017/05
Seattle Loop	8760	323	2015/01–2015/12
Weather	52,696	21	2020/01–2021/01
ECL	26,304	321	2012/01–2014/01

(Q5) How robust is the performance of DFE when faced with common data imperfections such as noise and missing values?

(Q6) Are the performance improvements offered by DFE stable and consistent across multiple independent trials?

III. In-depth Analysis and Ablation Studies: Next, we delve into the mechanics of our approach to understand why it works and to justify our design choices.

(Q7) How prevalent is the dormant node phenomenon in real-world traffic datasets, and does our method effectively target it?

(Q8) Do experimental results validate DFE's effectiveness in addressing Dormant Nodes beyond theoretical analysis?

(Q9) Is the architectural design of DFE, particularly its learnable nature, well-justified compared to simpler alternatives?

(Q10) Are there other potential strategies for addressing the dormant node problem, and how do they compare to our feature augmentation approach?

IV. Generality and Problem Validation: Finally, we examine the broader applicability of our method and the problem itself.

(Q11) Is DFE effective in other domains, extending beyond traffic forecasting to other multivariate time series tasks?

Through answering these questions, we aim to provide a thorough and convincing demonstration of our contributions.

6.1. Experimental settings

This subsection consolidates all experimental ingredients required for reproducibility. Section 6.1.1 describes every dataset in detail, Section 6.1.2 lists the competing baselines, and Section 6.1.3 specifies the training protocol and evaluation metrics.

6.1.1. Datasets

To ensure a comprehensive evaluation, we utilize a diverse set of nine real-world datasets, categorized into two groups. The first group consists of seven widely-used traffic forecasting datasets: **PEMS-03**, **PEMS-04**, **PEMS-07**, **PEMS-08**, **METR-LA**, **PEMS-BAY**, and **Seattle Loop**. The second group includes two general multivariate time series datasets, **Weather** and **Electricity (ECL)**, to validate the broader applicability and generalizability of our proposed method. Table 1 summarizes the key statistics of all datasets.

1. **PEMS-03:** Contains traffic flow data from 358 sensors in Caltrans District 3, recorded every five minutes.
2. **PEMS-04:** Includes traffic data from 307 sensors in the San Francisco Bay Area, collected at five-minute intervals.
3. **PEMS-07:** Comprises data from 883 sensors in the Los Angeles area at five-minute intervals.
4. **PEMS-08:** Includes data from 170 sensors in the San Bernardino area, recorded every five minutes.

5. **METR-LA:** A traffic speed dataset from 207 sensors in Los Angeles County, recorded every five minutes.
6. **PEMS-BAY:** Contains traffic speed data from 325 sensors in the San Francisco Bay Area, collected every five minutes.
7. **Seattle Loop:** Provides hourly traffic flow data from 323 detectors across the Seattle area.
8. **Weather:** Contains 21 meteorological indicators (e.g., temperature, humidity) recorded every 10 min.
9. **Electricity (ECL):** Records the hourly electricity consumption of 321 clients.

6.1.2. Baselines

We evaluate the performance of DFE against a wide range of baseline models, which can be grouped into three categories:

1. Traditional Time Series Models:

- **HA** (Box et al., 2015): Historical Average, a simple benchmark that predicts future values by averaging historical data.
- **VAR** (Stock and Watson, 2001): Vector AutoRegression, a classical model that captures linear interdependencies across multiple time series.

2. Graph-based Traffic Forecasting Models:

- **DCRNN** (Li et al., 2018): Integrates diffusion convolution with recurrent neural networks.
- **STGCN** (Yu et al., 2018): Employs graph and temporal convolutions to model spatio-temporal dependencies.
- **ASTGCN** (Guo et al., 2019): Extends STGCN with attention mechanisms.
- **GWNet** (Wu et al., 2019): Combines dilated causal convolutions with graph convolutions for dynamic correlations.
- **GMAN** (Zheng et al., 2020): Utilizes spatial and temporal attention mechanisms to capture complex dependencies.
- **AGCRN** (Bai et al., 2020): Learns node-specific patterns through adaptive graph structures.
- **DMSTGCN** (Han et al., 2021): Employs multiple spatial learning strategies for richer feature representations.
- **SSTBAN** (Guo et al., 2023): Introduces a bottleneck attention design for computational efficiency.
- **STPGNN** (Kong et al., 2024): Proposes a spatio-temporal position-aware GNN to capture positional information.
- **Extralonger** (Z. Zhang et al., 2024): A recent SOTA model unifying spatial and temporal factors for long-term forecasting.

3. General-Purpose Forecasting Models:

- **DLinear** (Zeng et al., 2023): A simple yet powerful linear model that serves as a strong baseline against complex models.
- **iTransformer** (Liu et al., 2024): A recent Transformer variant that applies attention to the feature dimension for time series representation.

Furthermore, the non-learnable Dormant Feature Indicator (DFI) is used as a minimalistic ablation baseline to directly validate the effectiveness of dormant feature augmentation.

6.1.3. Implementation details and evaluation metrics

We evaluate the effectiveness of our proposed Dormant Feature Embedding (DFE) by integrating it into the aforementioned baseline models. To isolate the impact of DFE, we apply it as an additional feature channel at the input layer, keeping all other components and hyperparameters identical to the original open-source implementations.

Table 2

Comparison of baseline models and their DFI- and DFE-enhanced variants. The best results are in bold. Results for STPGNN on the Seattle Loop dataset are omitted, as the model exhibited unstable performance and failed to converge in our experiments.

Dataset	Method	24 time steps			36 time steps			48 time steps		
		RMSE	MAE	MAPE	RMSE	MAE	MAPE	RMSE	MAE	MAPE
PEMS04	HA	81.57	56.47	45.49	106.58	76.01	68.84	127.28	93.37	94.62
	DCRNN	42.86	28.70	21.23	51.40	33.78	27.10	57.85	38.26	33.73
	GWNet	35.52	22.79	16.04	38.17	24.71	17.67	40.60	26.42	18.99
	GMAN	38.10	21.67	17.78	52.86	22.12	16.43	47.85	23.35	17.98
	AGCRN	34.44	21.63	14.65	38.19	24.15	16.33	38.26	24.18	16.31
	DMSTGCN	32.09	20.32	14.13	34.86	22.47	15.86	35.05	22.50	16.56
	DMSTGCN+DFE	31.70	19.99	13.81	34.22	22.01	15.40	34.68	22.19	15.99
	SSTBAN	33.15	20.36	14.73	34.15	20.82	14.83	35.81	21.66	15.90
	SSTBAN+DFI	32.74	20.18	14.69	33.89	20.61	14.89	35.63	21.61	15.30
	SSTBAN+DFE	32.57	19.59	13.71	33.79	<u>20.21</u>	14.20	34.80	21.17	14.87
	STPGNN	31.84	19.55	13.69	33.91	20.82	14.45	35.03	21.65	15.29
	STPGNN+DFI	31.56	19.32	13.54	33.54	20.57	14.24	34.55	21.35	15.08
	STPGNN+DFE	31.25	19.04	13.43	33.09	20.48	14.13	34.36	21.18	14.89
	Extralonger	32.01	19.69	13.60	33.52	20.58	14.13	34.41	21.15	14.65
PEMS08	Extralonger+DFI	31.66	19.61	13.32	32.85	20.29	13.77	33.78	20.83	14.49
	Extralonger+DFE	31.42	19.40	13.23	32.84	20.16	13.65	33.66	20.65	14.17
	HA	69.72	48.30	32.09	92.72	65.99	46.64	111.85	81.51	61.29
	DCRNN	33.34	22.60	15.46	39.37	25.82	18.53	45.64	30.47	25.10
	GWNet	29.47	19.07	12.25	33.54	21.76	13.68	34.20	22.60	14.16
	GMAN	34.29	17.38	15.66	35.89	17.21	16.33	48.54	18.70	16.81
	AGCRN	28.05	17.45	11.25	30.96	19.39	12.73	31.11	19.46	12.88
	DMSTGCN	26.55	16.75	11.44	28.50	18.15	12.64	28.94	18.34	12.93
	DMSTGCN+DFE	26.07	16.41	11.12	27.98	17.71	12.35	28.53	17.94	12.46
	SSTBAN	26.32	15.97	12.29	28.30	16.84	12.20	28.82	16.94	12.47
	SSTBAN+DFI	25.96	15.74	11.55	<u>27.58</u>	16.37	11.32	28.66	16.85	11.94
	SSTBAN+DFE	<u>25.91</u>	<u>15.51</u>	11.12	27.08	16.34	11.57	28.62	16.63	11.88
	STPGNN	26.12	16.05	10.31	28.01	16.99	11.61	28.95	17.62	11.90
	STPGNN+DFI	25.81	15.84	10.15	27.72	16.73	11.41	28.68	17.35	11.76
Seattle Loop	STPGNN+DFE	25.60	15.62	10.11	27.46	16.64	11.31	28.51	17.30	11.65
	Extralonger	26.39	15.89	10.49	28.10	16.72	11.36	29.01	17.31	11.66
	Extralonger+DFI	26.08	15.64	<u>10.29</u>	27.61	16.40	<u>10.81</u>	<u>28.59</u>	16.93	<u>11.38</u>
	Extralonger+DFE	25.79	15.41	10.16	27.59	<u>16.37</u>	10.80	28.49	<u>16.85</u>	11.21
	HA	11.86	8.07	26.57	12.37	8.47	27.76	12.31	8.49	27.82
	VAR	9.56	6.21	19.94	9.96	6.44	21.28	10.28	6.69	22.24
	DCRNN	7.97	4.37	14.04	8.38	4.60	14.41	8.63	4.73	14.91
	GWNet	7.84	4.28	14.06	8.18	4.60	14.92	8.35	4.67	15.04
	GMAN	7.84	4.13	12.88	8.10	4.23	12.95	8.09	4.26	13.26
	AGCRN	7.83	4.27	13.53	8.31	4.64	14.17	8.48	4.75	15.02
	DMSTGCN	7.59	4.08	13.51	7.98	4.31	14.31	8.20	4.49	14.86
	DMSTGCN+DFE	7.51	3.98	13.17	7.82	4.27	14.11	8.09	4.38	14.30
	SSTBAN	7.72	4.05	12.69	7.83	4.11	12.44	7.88	4.12	12.35
	SSTBAN+DFI	7.62	4.00	12.48	7.72	4.06	12.26	7.77	4.06	12.16
	SSTBAN+DFE	7.57	<u>3.97</u>	12.38	7.67	4.04	12.17	7.72	<u>4.03</u>	12.06
Seattle Loop	Extralonger	7.43	4.04	12.48	7.51	4.05	11.96	7.68	4.11	12.04
	Extralonger+DFI	7.25	3.99	12.15	<u>7.45</u>	4.01	<u>11.89</u>	<u>7.61</u>	4.09	<u>11.99</u>
	Extralonger+DFE	7.15	3.88	11.80	7.42	3.99	11.80	7.58	4.06	11.85

All experiments are performed with multiple independent repetitions to ensure statistical robustness.

Experimental Environment. All experiments are conducted using PyTorch on a workstation equipped with an NVIDIA RTX 4090D GPU and an Intel Xeon(R) Gold 6430 CPU. We strictly adhere to the original baselines' hyperparameter configurations to ensure a fair comparison. We assess performance over three categories of forecasting horizons: Short-Term (12 steps, e.g., 1 h), Long-Term (48 steps, e.g., 4 h), and Extra-Long-Term (up to 2016 steps, e.g., 7 days).

Evaluation Metrics. To comprehensively evaluate forecasting accuracy, we adopt four complementary metrics: Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and Weighted MAPE (WMAPE).

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}, \quad \text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|, \quad (10a)$$

$$\text{MAPE} = \frac{1}{N} \sum_{i=1}^N \left| \frac{y_i - \hat{y}_i}{y_i} \right|, \quad \text{WMAPE} = \frac{\sum_{i=1}^N |y_i - \hat{y}_i|}{\sum_{i=1}^N y_i}. \quad (10b)$$

6.2. Long-term forecasting performance (Q1)

To comprehensively evaluate the effectiveness of our proposed dormant feature methods in long-term forecasting, we integrate DFE and its simplified ablation baseline, DFI, into several advanced baseline models. We conduct tests on three standard datasets, PEMS04, PEMS08, and Seattle Loop, with detailed performance comparisons presented in Table 2.

The experimental results clearly indicate that introducing dormant features consistently enhances prediction performance. For instance, on the PEMS04 dataset, applying DFE to the SSTBAN model significantly reduces its MAPE by 6.9%, 4.2%, and 6.5% across the three forecasting horizons, respectively. Similarly, on the PEMS08 dataset, DFE delivers an average MAPE reduction of 4.0% for the SOTA model Extralonger. These consistent improvements across different models and datasets validate the effectiveness of our approach.

Furthermore, a comparison between DFI and DFE reveals that while the simple binary indicator (DFI) already provides tangible performance gains, our proposed learnable embedding (DFE) achieves supe-

Table 3
Performance comparison in extra-long-term scenarios. TS: Length of time steps.

Dataset	TS	Extralonger			Extralonger+DFI			Extralonger+DFE		
		RMSE	MAE	MAPE	RMSE	MAE	MAPE	RMSE	MAE	MAPE
PEMS04	64	35.11	21.67	14.85	34.86	21.24	14.62	34.24	21.02	14.57
	144	39.33	23.66	16.04	38.62	23.43	15.88	37.96	22.98	15.73
	288	40.46	24.18	16.70	40.07	23.89	16.61	39.51	23.50	16.44
	576	42.15	26.08	19.32	40.17	24.25	16.80	39.83	24.04	16.42
	864	42.98	26.60	19.36	41.55	24.99	17.25	41.17	24.66	16.31
	1152	42.98	26.88	19.33	42.11	25.25	16.99	41.43	24.79	16.43
	1440	43.00	26.62	19.24	42.31	25.99	16.84	42.03	24.61	16.63
	1728	43.33	27.09	19.63	43.05	25.70	16.79	42.43	25.20	16.69
PEMS08	2016	43.26	27.52	20.21	42.55	25.82	16.88	41.07	24.81	16.04
	64	30.01	17.91	12.32	29.71	17.42	11.82	28.92	17.19	11.50
	144	32.17	18.84	12.85	31.99	18.54	12.57	31.64	18.40	12.39
	288	33.76	19.38	13.05	32.12	18.79	12.87	32.48	18.94	12.68
	576	37.73	22.06	14.23	35.68	20.84	13.63	34.23	19.80	13.15
	864	38.57	22.57	15.04	37.37	21.08	13.82	34.89	19.80	12.97
	1152	40.45	23.50	16.03	38.64	21.35	14.01	37.06	20.70	13.27
	1440	41.44	23.73	17.90	38.55	22.01	14.23	38.01	21.02	14.01
	1728	40.85	23.68	18.42	38.62	20.98	14.81	38.57	20.94	15.04
	2016	41.38	23.69	19.18	39.14	20.73	15.68	37.82	20.37	15.66

rior results in nearly all cases. This confirms the importance of adaptively learning the degree of node dormancy. In summary, these findings strongly support our core hypothesis: explicitly modeling dormant node information is an effective and reliable strategy for improving long-term forecasting performance.

6.3. Extra-long-term prediction performance (Q2)

To rigorously test DFE's capabilities under the most demanding conditions, we evaluate its performance in ultra-long-term forecasting—a scenario where minor prediction errors can compound catastrophically over time. We integrate DFE into Extralonger, the current state-of-the-art model specifically designed for horizons up to 2016 steps (equivalent to one week). The evaluation is conducted on the **PEMS04** and **PEMS08** datasets, with the static DFI serving as an ablation baseline.

The results, detailed in Table 3, show that explicitly modeling dormancy is crucial for long-horizon stability. The DFE-enhanced model consistently and significantly outperforms the baseline across all metrics and time steps. For example, on PEMS08, the average reductions in RMSE, MAE, and MAPE are 6.53%, 8.90%, and 12.23%, respectively. Crucially, the performance gains amplify as the prediction horizon extends; the MAPE improvement on PEMS08 grows from a solid 6.65% at 64 steps to a substantial 18.35% at 2016 steps.

This trend confirms that DFE is highly effective at mitigating long-horizon error accumulation. In auto-regressive forecasting, the model's flawed understanding of dormant nodes leads to small initial errors. These errors are then fed back into the model, causing a cascade of compounding inaccuracies. By providing a stable “guiding signal”, DFE helps the model form a more robust initial representation of the entire network's dynamics, thus minimizing these initial errors and preventing their catastrophic amplification. Furthermore, the superiority of the learnable DFE over the static DFI highlights the importance of task-specific adaptation; the model learns the optimal dormancy representation needed to stabilize its long-term rollouts.

6.4. Short-term forecasting potential of DFE (q3)

While DFE is primarily designed to combat long-term error accumulation, we also investigate its utility in standard short-term forecasting. Our hypothesis is that the dormant node problem is not merely about error compounding, but is a more fundamental issue of representation learning. Even for single-step predictions, the corrupted, information-poor signals from dormant nodes can pollute the feature aggregation process, leading to less accurate spatial representations for all nodes in their vicinity.

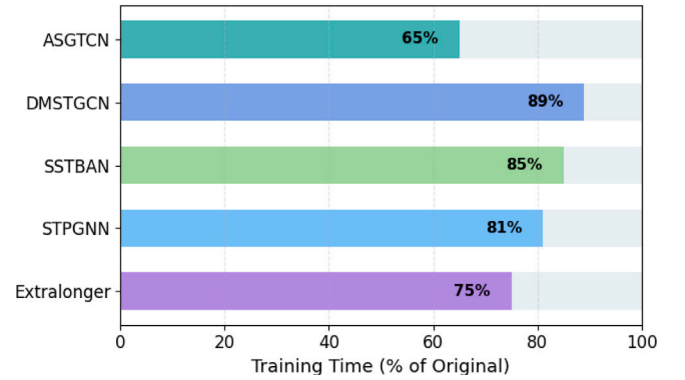


Fig. 3. Training time reduction across models after incorporating DFE (in seconds).

To test this, we integrate DFE into two foundational graph-based models, STGCN and ASGTCN, and evaluate them on four datasets. As shown in Table 4, DFE consistently improves short-term accuracy across all settings. On average, it reduces RMSE by 3.84%, MAE by 4.96%, and MAPE/WMAPE by 6.85%. These consistent gains, although more modest than in the long-term setting, are significant. They demonstrate that by helping the model correctly interpret or down-weight unreliable signals, DFE facilitates the learning of cleaner and more robust spatio-temporal representations from the outset. This validates that the dormant node problem is a fundamental challenge in spatio-temporal graph learning, and DFE is a generally applicable solution, not one limited to long-horizon scenarios.

6.5. Resource consumption analysis (Q4)

To evaluate the computational impact of DFE, we assess its effect on memory usage, inference latency, and overall training time. As a lightweight, input-level module, DFE introduces a negligible number of learnable parameters (one per node) and does not alter the downstream model architecture. Consequently, its impact on memory footprint and inference latency is virtually zero, preserving the efficiency of the original models.

Interestingly, while adding a component, DFE consistently improves training efficiency. As shown in Fig. 3, which presents the training time of DFE-enhanced models as a percentage of their original counterparts, all tested models exhibit a noticeable reduction in training duration. For instance, the state-of-the-art *Extralonger* model's training time is

Table 4
Short-term forecasting performance of STGCN and ASTGCN with and without DFE.

Model	Dataset	6 time steps			12 time steps		
		RMSE	MAE	MAPE/WMAPE	RMSE	MAE	MAPE/WMAPE
STGCN	METR-LA	11.84	6.51	0.128	14.55	8.21	0.162
STGCN+DFE	METR-LA	11.60	6.35	0.124	14.25	7.76	0.149
STGCN	PEMS-BAY	4.16	2.10	0.034	4.70	2.25	0.0367
STGCN+DFE	PEMS-BAY	4.09	2.01	0.032	4.55	2.19	0.0352
ASTGCN	PEMS04	37.46	23.94	16.12	35.31	22.52	15.36
ASTGCN+DFE	PEMS04	35.43	22.53	14.63	33.51	21.01	13.89
ASTGCN	PEMS08	30.74	20.06	12.33	28.91	18.59	11.45
ASTGCN+DFE	PEMS08	28.55	18.68	11.05	27.45	17.81	10.93

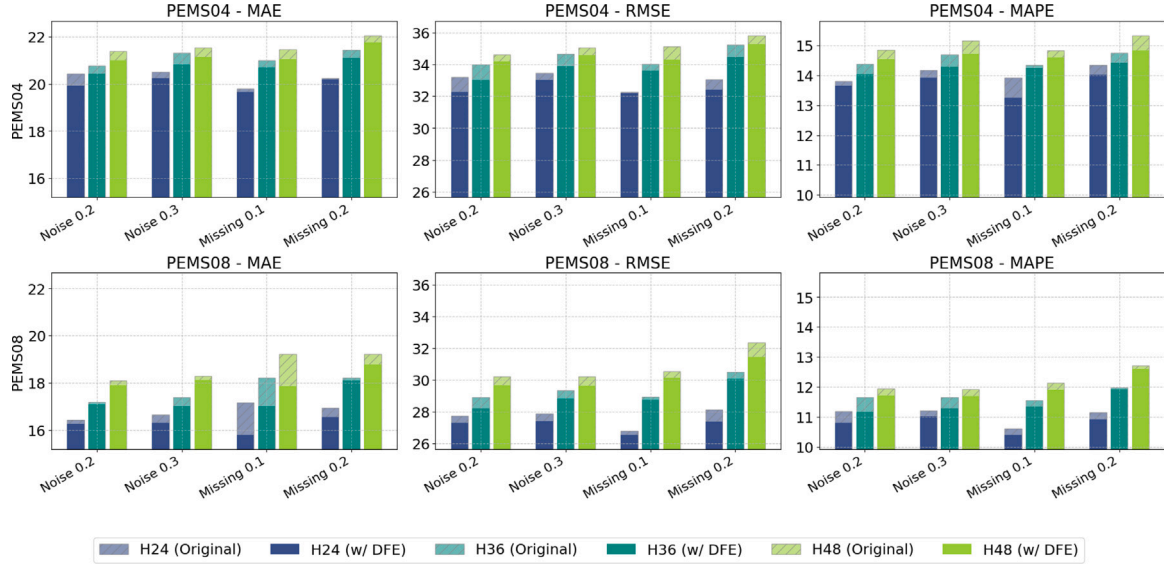


Fig. 4. DFE improves robustness. Darker bars represent DFE-enhanced models across different horizons and disturbance types.

reduced to approximately 75% of its baseline, with some models like *DMSTGCN* seeing an even greater reduction to 65%. This acceleration stems from faster convergence. By providing a clear and stable “guiding signal” about node dormancy, DFE helps the model quickly learn to handle unreliable, information-poor inputs instead of struggling to interpret noise. This makes the optimization process more stable and direct, allowing the model to reach its optimal state in fewer epochs. The practical benefit is significant, as faster training cycles enable more rapid experimentation and hyperparameter tuning.

6.6. Evaluation of DFE robustness under adverse real-world conditions (Q5)

Real-world traffic data is often imperfect, subject to sensor noise and data transmission failures. Therefore, we evaluate DFE’s robustness by simulating two common types of disturbances on the PEMS04 and PEMS08 datasets: (1) Gaussian noise (std = 0.2 and 0.3) and (2) missing data (10% and 20% random loss). Experiments are conducted using the *Extralonger* model across 24-, 36-, and 48-step prediction horizons.

The results, presented in Fig. 4, demonstrate that DFE provides a consistent and significant performance advantage across all adverse conditions. The figure details a comparison for each metric (MAE, RMSE, MAPE); under every disturbance scenario, the DFE-enhanced model (darker bars) systematically outperforms its baseline counterpart (lighter bars). This indicates that DFE’s benefits are not confined to clean data but are maintained, and sometimes even amplified, when data quality degrades.

This strong resilience can be explained by our theoretical framework. The core problem of dormant nodes is already a low signal-to-noise ratio.

- **Against Noise:** When external noise is added, the primary traffic signal becomes even more corrupted. DFE, however, provides a clean, noise-free side-channel of information (the dormancy flag). The model can learn to rely more on this stable DFE signal to correctly interpret the context of the noisy traffic data, leading to more robust predictions.

- **Against Missing Data:** When a dormant node’s data is missing, the loss of dynamic information is minimal. However, DFE continues to provide crucial *static* information: the node’s existence and its typical low-activity characteristic. This helps the model preserve its understanding of the overall graph topology, even when parts of the dynamic input are absent.

This demonstrates that DFE is not just an accuracy booster but also a practical stabilization module, making spatio-temporal models more reliable for real-world deployment.

6.7. Stability analysis (Q6)

To further validate the robustness and consistency of our method, we present an error bar comparison in Fig. 5. Specifically, we choose *Extralonger*, the current state-of-the-art model for long-horizon forecasting, and evaluate it on the **PEMS07** dataset, where our proposed DFE module demonstrates the most significant gains. We conduct experiments across three time horizons: 24, 36, and 48 steps, and visualize the distribution of RMSE, MAE, and MAPE over multiple runs. The results clearly show that *Extralonger*+DFE not only achieves lower average error across all metrics but also exhibits smaller variance, indicating improved stability and generalization.

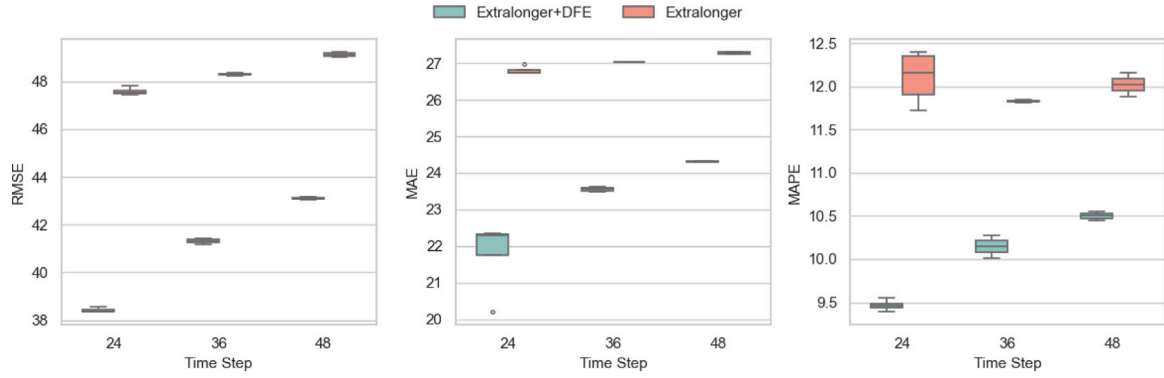


Fig. 5. Error bar comparison of Extralonger with and without DFE on PEMS07 across three time steps.

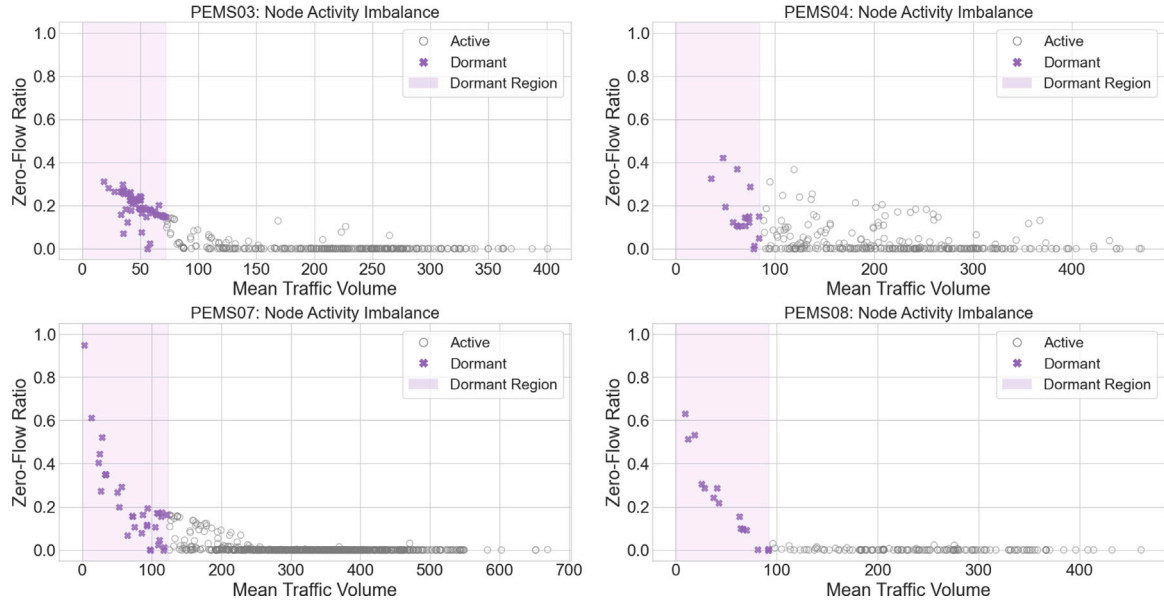


Fig. 6. Dormant node distribution across four traffic datasets. The shaded region indicates the dormant zone, defined as nodes with mean traffic below 40% of the global average. Nodes are categorized and styled for clarity: dormant nodes (purple 'X') fall within the shaded region, while active nodes (gray hollow circles) lie outside of it. Subplots correspond to PEMS03, PEMS04, PEMS07, and PEMS08 (top-left to bottom-right).

6.8. Characterizing the dormant node phenomenon (Q7)

To assess the prevalence and severity of dormant nodes in real-world traffic networks, we first analyze four benchmark datasets. A node is considered *dormant* if its average traffic falls below 40% of the global mean; a *zero-flow state* occurs when traffic drops below 5% of the mean. Note that while the bottom 5% percentile is used for the algorithmic implementation of DFE to ensure cross-dataset consistency, the 40% global mean ratio is adopted here solely as a descriptive metric to characterize and compare the intrinsic dormancy severity across different urban networks. As shown in Fig. 6, all datasets exhibit dormant-node patterns with varying intensity: PEMS03 shows widespread intermittent inactivity; PEMS04 and PEMS08 contain fewer dormant nodes (though PEMS08 has some with prolonged near-zero flow); and PEMS07 is the most extreme, with several nodes inactive for nearly the entire year, forming persistent “dead zones”.

To quantify the severity of this phenomenon, we define a Dormancy Severity Index (DSI) as $DSI = DNR \times (ZFR)^2 \times N$, where DNR is the ratio of dormant nodes, ZFR is their average zero-flow ratio, and N is the total number of nodes. Squaring ZFR emphasizes the persistence of zero-flow states. Fig. 8 visualizes the DSI for several datasets, confirming that PEMS07 exhibits the most severe dormancy, followed by PEMS08, while PEMS04 is relatively mild.

Table 5

Performance on PEMS07 using Extralonger with/without DFE.

Horizon	Extralonger			Extralonger+DFE		
	RMSE	MAE	MAPE	RMSE	MAE	MAPE
24	47.48	26.74	12.35	38.40	22.25	9.45
36	48.50	27.03	11.85	41.20	23.50	10.11
48	49.24	27.25	12.06	43.08	24.31	10.50

6.9. Empirical validation of DFE’s effectiveness (Q8)

Having established the prevalence and severity of the dormant node phenomenon, we now empirically validate the effectiveness of our proposed Dormant Feature Embedding (DFE) in mitigating its effects. A case study on PEMS07 — the dataset with the highest DSI — using the Extralonger model (Table 5) shows that DFE yields dramatic improvements across all metrics under these conditions of extreme dormancy.

To generalize this finding, we correlate each dataset’s DSI with the relative performance gains achieved by DFE. As illustrated in Fig. 7, there is a clear positive correlation between a dataset’s DSI and the error reduction in RMSE, MAE, and MAPE. This strongly confirms our

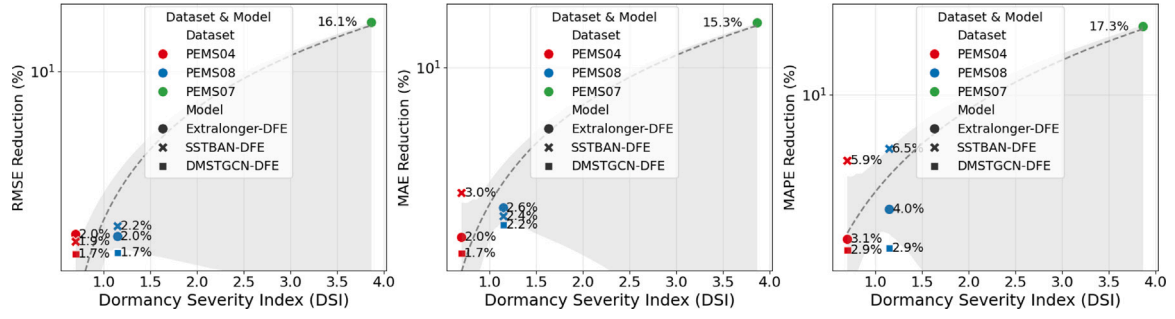


Fig. 7. Relationship between Dormancy Severity Index (DSI) and the reduction percentages of RMSE, MAE, and MAPE due to the inclusion of Dormant Feature Embedding (DFE).

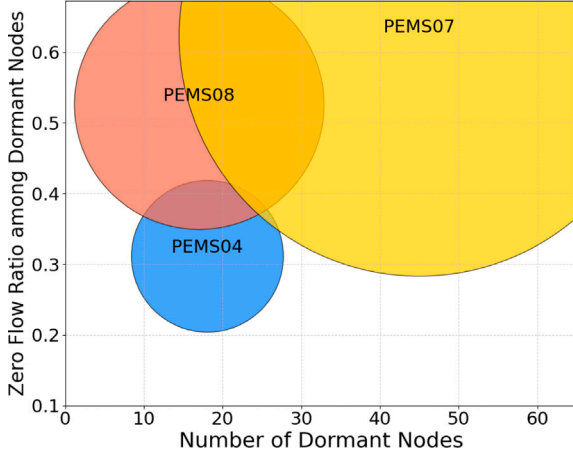


Fig. 8. Bubble size indicates Dormancy Severity Index.

Table 6

Quantitative analysis of latent feature standard deviation (Std) with and without DFE.

Dataset	Model	All-Feature Avg Std	Relative increase
PEMS04	Extralonger	0.3719	+1.58%
	Extralonger+DFE	0.3778	
PEMS08	Extralonger	0.3599	+3.55%
	Extralonger+DFE	0.3727	

central hypothesis: DFE provides greater benefits precisely where the dormancy problem is most severe.

To quantitatively validate our information-theoretic propositions (as discussed in Section 5) and to verify that DFE practically restores the information entropy of dormant nodes, we conducted an intermediate feature analysis. In representation learning, the variance or standard deviation (Std) of intermediate hidden features serves as a robust empirical proxy for information entropy. A near-zero variance indicates representation collapse (i.e., a “dead” signal corrupted by noise), whereas a healthy variance indicates rich information capacity.

We extracted the high-dimensional latent spatial representations immediately preceding the spatial graph convolution layers of the *Extralonger* model. We then calculated the average feature standard deviation across all nodes on both the PEMS04 and PEMS08 datasets. The results are summarized in Table 6.

As shown in Table 6, the injection of DFE consistently increases the overall feature standard deviation without triggering gradient instability. This provides direct quantitative evidence that DFE successfully “uncompresses” the signals of dormant nodes, restoring their representational magnitude in the latent space and preventing them from being overwhelmed by global normalization.

Table 7

Ablation study on PEMS04, 36-step horizon.

Setting	RMSE	MAE	MAPE
Extra+Mix branch	33.05	20.28	13.79
STPGNN+Node-norm	37.11	23.25	15.98
Extra+Node-norm	36.53	22.84	15.79
STPGNN	33.91	20.82	14.45
Extralonger	33.52	20.58	14.13

Crucially, this analysis reveals a striking alignment with our previous macroscopic findings: the relative increase in feature variance on PEMS08 (+3.55%) is significantly larger than that on PEMS04 (+1.58%). This perfectly corroborates the positive correlation established in Fig. 7. Because PEMS08 suffers from a more severe dormancy phenomenon (a higher DSI), the corresponding entropy recovery mandated by DFE is mathematically more pronounced. This intermediate-level evidence elegantly bridges the gap between our theoretical motivation and the final predictive performance.

Finally, we performed a granularity analysis on PEMS04 by ranking nodes by their average traffic and grouping them into low, medium, and high-activity categories. As shown in Fig. 9, DFE produces the largest improvements for both the “coldest” (most dormant) and “hottest” (most active) nodes. This aligns with our intuition that the most dormant nodes benefit from direct feature enhancement, while the most active nodes benefit from the model’s improved understanding of the entire network, which is no longer being corrupted by noisy signals from dormant neighbors.

6.10. Alternative strategies (Q9)

Given that the core challenge in traffic forecasting stems from inherent aggregation in flow data — where a subset of nodes consistently exhibits low activity and thus severe imbalance — an intuitive remedy is per-node normalization, which should, in principle, remove scale disparities and preserve gradient signals otherwise compressed by global normalization.

To systematically evaluate normalization strategies, we conducted ablation studies with Extralonger and STPGNN, replacing global normalization with per-node normalization. As shown in Table 7, both models experience a clear drop in accuracy with node-wise normalization. This likely results from the strong spatial dependencies in traffic networks: node-wise normalization disrupts cross-node relationships, hindering the capture of complex spatial patterns.

We also explored a dual-normalization scheme, where globally and node-wise normalized features are processed in parallel by Extralonger and predictions fused at the output. This hybrid approach (Extra+Mix branch) improves accuracy over either strategy alone, but nearly doubles parameter count, making it impractical for long-term scenarios.

In contrast, our DFE-based feature augmentation is lightweight and efficient, achieving similar or better improvements with minimal

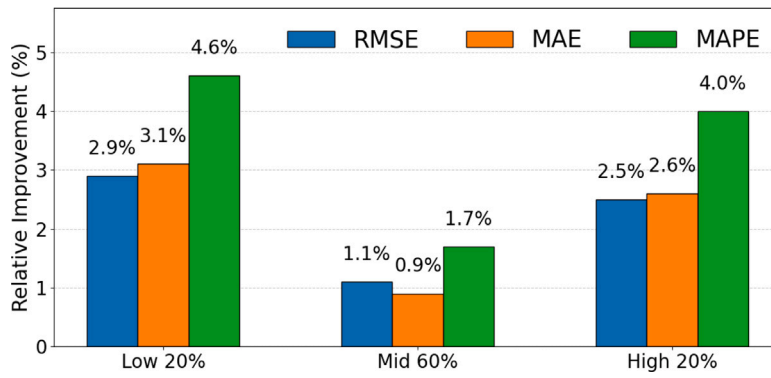


Fig. 9. Relative gain (%) of the Extralonger with DFE.

Table 8

Ablation study on PEMS04, 36-step horizon. Percentage values denote the average relative change compared to the base model.

Setting	RMSE	MAE	MAPE	Change (%)
SSTBAN	34.15	20.82	14.83	–
SSTBAN+LM	34.35	21.01	14.79	+0.41%
SSTBAN+DLM	33.95	20.67	14.95	–0.17%
SSTBAN+DFE (10%)	33.88	20.35	14.42	–1.94%
SSTBAN+DFE (Sigmoid)	33.81	20.28	14.25	–2.53%
SSTBAN+DFE (5%)	33.79	20.21	14.20	–2.74%
STPGNN	33.91	20.82	14.45	–
STPGNN+LM	33.95	20.77	14.49	+0.05%
STPGNN+DLM	33.44	20.58	14.33	–1.12%
STPGNN+DFE (10%)	33.28	20.58	14.27	–1.42%
STPGNN+DFE (Sigmoid)	33.14	20.55	14.17	–1.84%
STPGNN+DFE (5%)	33.09	20.48	14.13	–2.09%
Extralong	33.52	20.58	14.13	–
Extralong+LM	33.45	20.65	14.07	–0.10%
Extralong+DLM	32.89	20.14	13.61	–2.51%
Extralong+DFE (10%)	32.95	20.17	13.63	–2.41%
Extralong+DFE (Sigmoid)	32.91	20.20	13.69	–2.26%
Extralong+DFE (5%)	32.84	20.16	13.65	–2.49%

overhead. This underscores that targeted feature enhancement is a more practical and scalable solution to the dormant node problem in spatiotemporal forecasting.

6.11. Structure analysis (Q10)

To systematically address the dormant-node issue via feature enhancement, we designed and evaluated several variants of node-wise augmentation. The most basic strategy, the Learnable Mask (LM), assigns each node an unconstrained trainable scalar without any prior information. Building on this, the Data-informed Learnable Mask (DLM) initializes each scalar with prior knowledge: masks are set according to node activity, with the hottest node initialized to one and the coldest to zero. Beyond these two baselines, we further explored modifications to our proposed DFE structure, including replacing the Gumbel-Sigmoid activation with a standard Sigmoid (DFE (Sigmoid)) and relaxing the dormant-node threshold from 5% to 10% (DFE (10%)). All variants were evaluated under identical settings to ensure a fair comparison of their design choices.

The experimental results in Table 8 provide a clear picture of the relative effectiveness of different feature enhancement strategies. As expected, the LM variant — which introduces node-wise learnable masks without any prior knowledge — yields negligible improvement over the baseline, highlighting the limited benefit of unconstrained parameterization alone.

Introducing prior knowledge through DLM, which softly re-weights all nodes according to their activity level (from coldest to hottest), does

Table 9

Statistical analysis of DFE mask learnability (micro-calibration) for initialized dormant nodes.

Dataset	Micro-calibration direction		Mask variance (Std)	
	Enhanced (↑)	Suppressed (↓)	Initial	Learned
PEMS04	0 (0.0%)	16 (100.0%)	0.0	4.2×10^{-5}
PEMS08	7 (77.8%)	2 (22.2%)	0.0	7.3×10^{-5}

yield moderate gains. However, these gains are not consistent across models, as DLM relies heavily on the backbone’s capacity to exploit such global priors. In practice, this mild global re-weighting can be either helpful or suboptimal depending on how the model itself handles spatial dependencies and node-level heterogeneity.

In contrast, all three DFE variants — regardless of specific design — demonstrate clear and consistent improvements. By selectively focusing on the most dormant nodes, DFE provides targeted enhancement where the information gain is largest. The choice of the default 5% threshold is rooted in the statistical rationale of isolating the extreme tail of the distribution, ensuring that only the most persistently inactive nodes are highlighted.

Conducted by comparing DFE (5%) with the DFE (10%) variant, reveals that a narrower threshold is generally superior. While expanding the mask to 10% of nodes still outperforms the baseline, the gains are noticeably diminished compared to the 5% setting. This suggests that a broader threshold begins to introduce noise by over-compensating nodes that are low-traffic but still functionally active. Thus, the 5% threshold serves as an empirical “sweet spot” that balances coverage with precision. Comparing Gumbel-Sigmoid and standard Sigmoid variants reveals only slight differences in performance; the stochastic gating offered by Gumbel-Sigmoid may offer minor stability advantages, but both approaches are robust and deliver substantial benefits over simpler, more global schemes.

Discussion on Module Learnability. To further investigate the actual learnability of the DFE module — specifically, whether it genuinely refines the prior or merely preserves the initial heuristic — we tracked the evolution of the mask weights during training on the state-of-the-art (SOTA) *Extralonger* architecture. A statistical analysis of the weight divergence for the initialized dormant nodes is presented in Table 9.

As shown in Table 9, even when integrated into a highly optimized SOTA architecture, the module successfully utilizes the global heuristic as an optimal starting point while dynamically customizes the representation for each node. The variance of the mask weights diverged from strictly 0.0 at initialization to measurable values post-training, proving that the model learns fine-grained spatial heterogeneity rather than treating all dormant nodes identically.

Crucially, the direction of this micro-calibration aligns perfectly with the dataset’s dormancy severity (DSI). On PEMS08 (higher DSI), the model adaptively decided to further enhance the weights for 77.8%

Table 10

Cross-domain performance of iTransformer and DLinear with and without DFE. All values are reported as MSE/MAE.

Dataset	Model	Horizon 96	Horizon 192
ECL	iTransformer	0.148/0.240	0.162/0.253
	iTransformer+DFE	0.147/0.239	0.161/0.251
	DLinear	0.197/0.282	0.196/0.285
	DLinear+DFE	0.195/0.281	0.195/0.284
Weather	iTransformer	0.174/0.214	0.221/0.254
	iTransformer+DFE	0.173/0.212	0.219/0.253
	DLinear	0.196/0.255	0.237/0.296
	DLinear+DFE	0.195/0.254	0.235/0.294

of the dormant nodes. Conversely, on PEMS04 (lower DSI), the model autonomously suppressed the weights for all dormant nodes to prevent over-compensation. This confirms that DFE genuinely learns to refine the imperfect prior into an optimal, customized distribution driven by the dataset's intrinsic traffic patterns.

6.12. Cross-domain validation (Q11)

To test DFE's generality beyond traffic forecasting, we apply it to *DLinear* and *iTransformer* on two standard benchmarks: ECL and WEATHER. The gains are modest: 0.36% for DLinear and 0.47% for iTransformer on average.

We attribute this to the *channel-independent normalization* used in these datasets, which standardizes each variable separately. As a result, even low-activity variables retain their fluctuations and remain learnable, reducing the need for further enhancement. In contrast, traffic datasets adopt *global normalization* across nodes, which can flatten low-traffic inputs into near-constants, weakening gradient flow. In such cases, DFE helps amplify dormant-node signals and improve learning. To preliminarily validate this hypothesis, we conduct an auxiliary experiment on Extralonger, where both globally-normalized and node-wise-normalized data are provided to the model. The two outputs are inversely normalized and averaged during inference. This hybrid setup yields an average improvement of 2.4% across metrics. However, under this dual-normalization setting, the additional gain from applying DFE becomes marginal, which aligns with our assumption that the dormant-node phenomenon is primarily induced by global normalization. As reported in Table 10, the performance gains on ECL and Weather datasets are more modest compared to traffic scenarios. This disparity highlights that the 'dormant node problem' is particularly critical in domains like traffic forecasting where global normalization is prevalent and strong spatial dependencies allow information-poor signals to propagate and corrupt the learning process.

7. Conclusion

In this paper, we identified and addressed the *dormant node problem*—a pervasive yet frequently overlooked challenge in spatiotemporal traffic forecasting. We demonstrated that a combination of intrinsic traffic data characteristics and global normalization practices can degrade the signals from persistently low-activity nodes, turning them into information-poor inputs that corrupt the learning of network-wide patterns and degrade long-horizon prediction performance.

To mitigate this issue, we proposed an effective and lightweight strategy of input-level feature augmentation. We first established a baseline with the Dormant Feature Indicator (DFI), a simple binary mask, which confirmed that explicitly signaling dormancy is beneficial. Building upon this insight, we developed the Dormant Feature Embedding (DFE), an adaptive and learnable module that allows the model to fine-tune the importance of the dormancy signal for each node.

Our extensive experiments provide compelling evidence for our approach. DFE consistently and significantly improves long- and ultra-long-term forecasting accuracy across a wide range of models and datasets, substantially outperforming the static DFI. Furthermore, our analysis revealed that DFE is a more robust solution than alternatives like per-node normalization, which can disrupt crucial spatial dependencies. The limited performance gains on general time-series datasets, which employ channel-independent normalization, further reinforce our hypothesis that the dormant node problem is most critical in domains where global data scaling is prevalent.

Importantly, our cross-domain analysis reveals that the dormant node problem is intrinsically tied to the global normalization schemes necessitated by spatially coupled architectures. Therefore, rather than a general fix for all multivariate time series, DFE is best understood as a targeted and highly effective solution specifically designed for domains like traffic networks, where maintaining spatial interdependencies forces the compression of low-activity signals.

In conclusion, this work presents DFE as a simple, model-agnostic, and scalable solution to the dormant node problem. By directly enhancing the input features, DFE offers a practical and powerful tool that advances the state-of-the-art in long-term traffic forecasting.

CRedit authorship contribution statement

Peng Wang: Writing – review & editing, Supervision, Resources, Funding acquisition. **Hairui Sun:** Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Kanghua Hui:** Writing — review & editing, Supervision, Funding acquisition.

Declaration of Generative AI and AI-assisted Technologies in the Manuscript Preparation Process

During the preparation of this work, the author(s) used Google's large language model to improve the language, grammar, and readability of the manuscript. After using this tool, the author(s) reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The datasets used in this study are publicly available, as cited in the manuscript. The source code and detailed references for the datasets are available at: <https://github.com/FizzSHR/Awakening-Dormant-Nodes>.

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